



This apples to oranges comparison indicates the effectiveness of a cluster-based system. A 23-node Prerna is five times as fast as the dual-processor Xserve. The Xserve sells for about \$4,000 (about Rs 1,88,000). Prerna, with 23 nodes costing Rs 10,000 per node, comes for Rs 2,30,000

Each node of Prerna is powered by an Intel Pentium 4 processor at 1.8 GHz, on an 845 chipset-based motherboard, with 256 MB DDR RAM, a 40 GB, 7,200 rpm hard disk, over a 100 Mbps network. Currently Prerna has 23 such nodes and is expected to hit 50 soon

both these libraries, the applications we intended to run were not available," he added.

MPI? PVM? Reads like a job for the TLA (Three Letter Acronym) police. Abhinav explained, "PVM and MPI are parallel libraries. PVM (Parallel Virtual machine) and MPI (Message Passing Interface), are sets of libraries using which parallel programs are developed. MPI is widely used. Our problem is that the programs which we have, such as the fibre-optics simulation, are Windows-based. So, we need to run them using VMware or WineX. But these are not optimised for PVM and MPI. Here, customised software development is required. A MATLAB MPI toolkit is available, but then the cluster would have been very specific. We wanted a general cluster which could compute a whole lot of stuff from simulation to games."

Why did Team Prerna decide to forego the Scalability Hero of Linux Clustering, Beowulf, for a newcomer? The reasons ranged from flexibility to cost. Vibhore says, "With openMosix, we can run almost all Linux apps, except Java threads. It gave us sound performance. As far as scalability is concerned, yes, we can only add 50 nodes using openMosix. Beowulf can run in thousands, and scores over openMosix when a job is complex... but openMosix is free and easier to implement."

openMosix is a delicate balance between the range of applications possible and the speed at which a job is completed. Take a complex job, such as a nuclear fission simulation, which require "tons of complicated calculations". If this task is optimised for MPI/PVM, the performance will be great, but sticking to these libraries takes away the flexibility afforded.

“Girlfriends? [Smiles] Not at present, I think VB has one but does not disclose... but who is Prerna [Laughs]? Team Prerna, on passions of the female persuasion”

Looking Ahead

While travelling at faster-than-bullet speeds along the roads laid by Moore's Law, the semiconductor industry has hit the proverbial wall. Diminishing returns and migrating electrons have made shrinking processors an uneconomical strategy. It seems the future lies in multi-core processors.

Clusters and distributed computers already exist. By parallelising a task, a cluster offers supercomputer speeds at a fraction of the cost.

"The speed and portability [of Prerna] interested us," Vibhore tells us, adding, "We imagine clusters being used instead of expensive servers... end-users will use these services to fuel ultra-modern gadgets such as wearable computers. At present, clusters will find more penetration in institutes and universities in India."

Presently, Prerna is used for "number crunching problems, and rendering using Blender and POV-Ray." Vibhore discusses plans to outsource its power, "We can provide computational services to agencies, which have massive calculation requirements. We can also help companies migrate from Windows to Linux, and set up clusters."

Abhinav chips in with a view to tomorrow, "We intend to shift it to the R & D facility of our college. Prerna will be used for MATLAB and fibre-optics simulations and for SETI@Home!"

The success of Linux with Prerna is also the inspiration for a cheap hardware routing solution. When their college router got "useless ARP packets", they thought of creating a router of their own. Abhinav says, "The main node of the cluster can also act as a router. For this, Vibhore and I are looking for a good firewall. The next phase will be to make router from very cheap hardware. Our estimate is Rs 7,500. We are studying FREESCO to gain knowledge. The cheap box concept is based on embedded Linux."

A likely application for a distributed or a cluster-based computing system, such as the Grid initiative, is to harness the power afforded, to drive low-powered devices such as PDAs. Think of a handheld computer running *Doom 3* by drawing power from a cluster of thousands of PCs.

A far-fetched concept? "Clusters can bring supercomputing power to small laboratories," Abhinav tells us, adding, "There are endless benefits... (A cluster) can be mixed with a SIMPUTER, where the latter will draw all the power from a cluster, reducing the price of the SIMPUTER to within reach of the common man."

Apologies to Frost: clusters sound promising and perhaps even deep. But Team Prerna has promises to keep, and miles to go before they sleep, and miles to go before they sleep. ■

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Write in

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Buzzword of the MONTH

MMOs

With the release of Ragnarok Online, Massively Multiplayer Online

(MMO) games have been heralded to Indian shores. Currently enjoying a beta phase, Ragnarok will soon come to a cyber-café near

you. Meanwhile, find it with the August issue of Digit. Globally, online games are expected to grow to a \$1.1 billion market by 2008